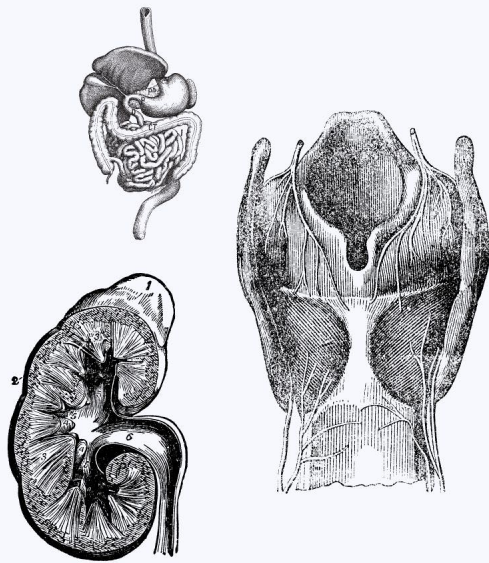




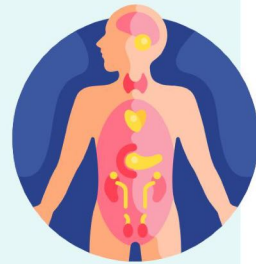
THE ENDOCRINE SYSTEM

H O R M O N E S .

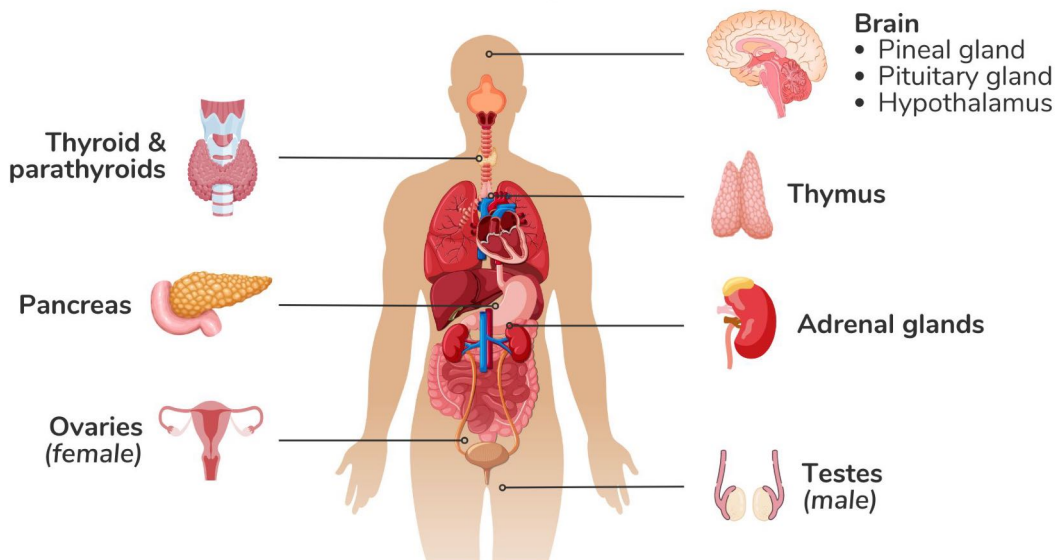
Part I: Anatomy + Physiology

Functions of the Endocrine System

- **Coordination of the male and female reproductive systems**
- **Maintenance of an optimal internal environment**
 - Keep the BG normal
 - Regulate normal metabolism levels
 - Regulate normal fluid levels
 - Regulate normal electrolyte levels
- **Respond to emergency demands**
 - Elevate BG in an emergency
 - Retain fluid to elevate the BP if hypotensive



Organs of the Endocrine System



Organ	Hormone
Hypothalamus	Thyroid Releasing Hormone (TRH) Corticotrophic-releasing hormone (CRH)
Pituitary Gland	Thyroid Stimulating Hormone (TSH) Folicle Stimulating Hormone (FSH) Lutenizing Hormone (LH) Oxytocin Antidiuretic Hormone (ADH, Vasopressin)
Parathyroid	Parathyroid Hormone (PTH)
Adrenal Glands	Steroids: glucocorticoids, mineralocorticoids
Pancreas	Insulin Glucagon
Thyroid Gland	T3, T4

Steroids

- Produced by the adrenal cortex
- Glucocorticoids
 - Affect mood
 - Cause immunosuppression
 - Breakdown fats & proteins
 - Inhibit insulin
- Mineralocorticoids - **aldosterone***
 - Retention of sodium and water
 - Excretion of potassium
- Sex hormones - **testosterone, estrogen, progesterone**

**Aldosterone connects the renal system to the endocrine system! RAAS system!*

Catecholamines

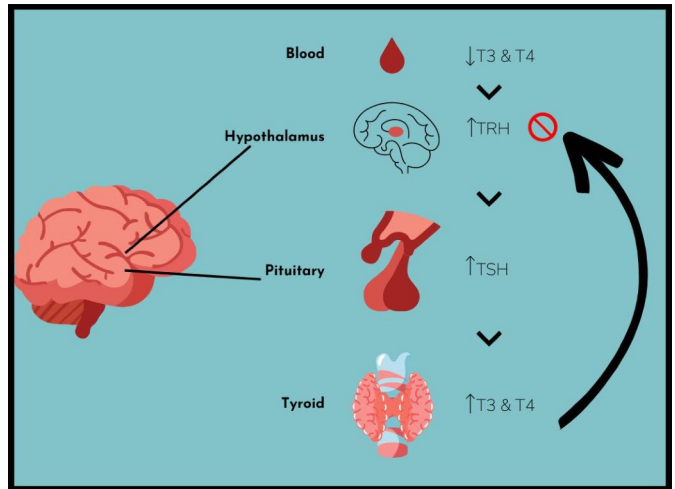
- Secreted by the adrenal *medulla*
- Norepinephrine
 - Vasoconstriction
 - Increased BP
 - Pupil dilation
 - Increased contractility
 - Inhibits GI system activity
- Epinephrine
 - Increases the body's metabolism
 - Increases glucose levels in the blood
 - Inhibits secretion of insulin

Antidiuretic Hormone

- Secreted from the **pituitary gland**
- Pituitary gland is in the brain, between your eyeballs
- Be on the lookout for these issues if a client had:
 - **Craniotomy**
 - Head injury
 - Sinus surgery
- Causes *anti* - diuresis - holding on to WATER
 - Only water is retained, so sodium!
 - Increased ADH → increased water
- Antidiuretic hormone = ADH = Vasopressin

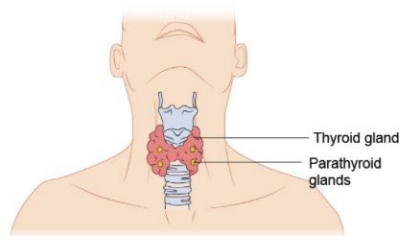
Thyroid hormone

- Produced by the thyroid gland
- There are two types: T3 and T4
- Thyroid hormones = energy
- They are controlled by a negative feedback loop
- Thyroid Stimulating Hormone (TSH) controls the release of T3 and T4
 - Low T3 and T4 cause High TSH
 - High T3 and T4 cause low TSH



Parathyroid Hormone

- Secreted by the parathyroid glands
- 4 parathyroid glands - located behind the thyroid gland.
- Causes calcium to be pulled out of the bones and into the blood.
- Causes an increase in serum calcium.

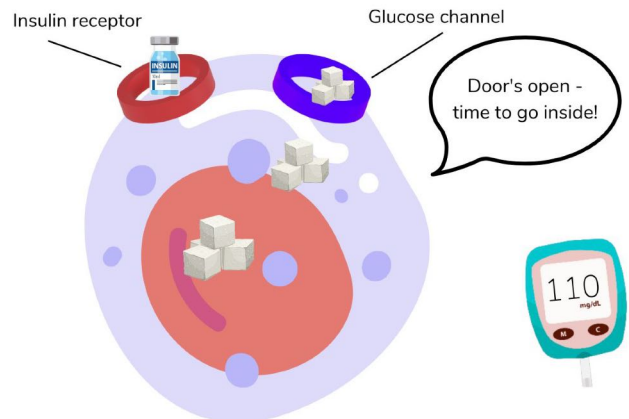


Calcitonin

- Produced by the thyroid gland
- Released when serum calcium levels increase
- Inhibits osteoclasts
 - Stops bones from releasing more calcium into the blood
 - Opposes PTH!
 - Protects against hypercalcemia!

Insulin

- Produced in the pancreas
 - β -islets of Langerhan
- Acts as the 'key' to transport glucose from the bloodstream to the cells
- Allows the cells to use glucose as fuel
- **Normal BG: 70-110**



Glucagon

- Produced by the α -cells of the pancreas
- Catabolic hormone
- Causes glycogenolysis in the liver
 - Stored glycogen $\rightarrow$ glucose
 - More glucose $\rightarrow$ raises blood glucose!
- Released when BG is low

Part II: Endocrine Pharmacology

Steroids

Short-Acting	• Cortisone • Hydrocortisone
Intermediate-Acting	• Methylprednisolone • Prednisone • Triamcinolone
Long-Acting	• Betamethasone • Dexamethasone

Methylprednisolone

- Therapeutic class: Corticosteroids
- Indication: Inflammation, allergy, autoimmune disorders
- Action: Suppress inflammation and normal immune response
- Nursing Considerations:
 - Monitor for too much steroids
 - Cushing's symptoms; buffalo hump
 - Side effects
 - Immunosuppression
 - Hyperglycemia
 - Osteoporosis
 - Delayed wound healing

Catecholamines

- Norepinephrine
 - Vasoconstriction
 - Increased BP
- Epinephrine
 - Inotrope

Antidiuretic Hormone

- Vasopressin
- Desmopressin

Levothyroxine

Indication: Hypothyroidism

Action: Replacement T4

Nursing Considerations:

- Must be taken on an empty stomach
- Take at the same time every day
- Lifelong therapy

Antithyroid Medications

- Iodine (Lugol's)
 - Reduces size and vascularity of the thyroid gland
- Radioactive iodide
 - Kills thyroid cells
- Methimazole
 - Stops the thyroid from producing T3 and T4

Parathyroid Medications

- Increase serum calcium
 - Calcium chloride, gluconate, lactate
 - Vitamin D
 - Calcitriol

- Decrease serum calcium
 - Bisphosphonates
 - Alendronate
 - Risedronate
 - Calcitonin

Insulin Onset, Peak, & Duration Times



Type	Generic Name	Onset	Peak	Duration
Rapid-Acting	Insulin aspart	15 min	1-3 hrs	3-5 hrs
	Insulin lispro	15 min	30-90 min	5 hrs
Short-Acting	Regular	30 min	2-4 hrs	5-12 hrs
Intermediate-Acting	NPH	1-4 hrs	4-12 hrs	10-24+ hrs
Long-Acting	Glargine	2-4 hrs	None	24 hrs



Insulin Storage

- Keep away from heat and direct sunlight
- Never freeze insulin
- Store in the refrigerator until ready for use
- When actively using, keep at room temperature
- At room temperature:
 - NPH: good for one month
 - Glargine: good for 28 days
 - Rapid and short acting: good for 28 days



Mixing Insulin

1. Draw up air equal to the total amount of insulin needed
2. Inject the correct amount of air into the NPH vial
3. Inject the remaining air into the regular insulin
4. Draw up the correct amount of regular insulin
5. Draw up the correct amount of NPH insulin

Clear, then cloudy (regular 1st, NPH 2nd!)

NEVER mix long-acting insulin

Mixing Insulin Memory Trick
"RN"



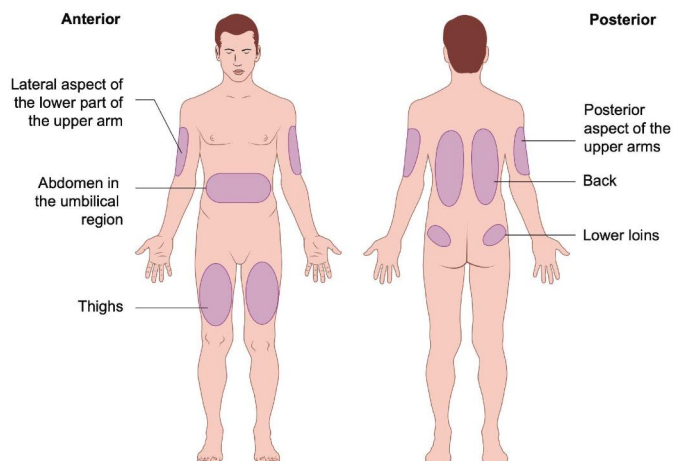
Draw up **Regular** ^{clear}
insulin first,
then draw up **NPH** _{cloudy}

Insulin Administration

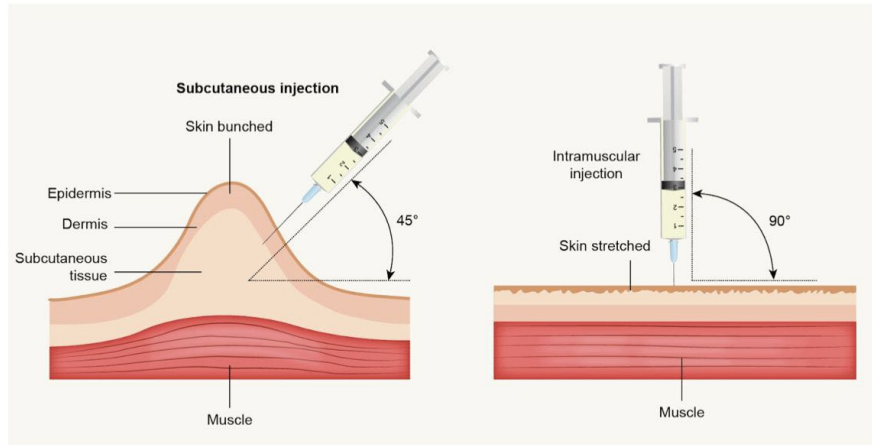
- Can only give Regular via IV
- All others given SubQ
- Rotate sites
- Syringes measured in units
- Never use expired or cloudy insulin
 - NPH is the only cloudy insulin



SubQ sites



Angle of Injection



Glucagon

- Therapeutic class: Catabolic hormone
- Indication: Emergency hypoglycemia treatment
- Action: Stimulate the liver to break down glycogen
- Nursing Considerations:
 - Administered IM
 - Teach client's how to administer properly!
 - Monitor BG

Antidiabetics

- Sulfonylureas
 - Glimepiride
 - Glyburide
- Biguanides
 - Metformin
- Insulin sensitizers
 - Rosiglitazone
 - Pioglitazone

Metformin

Therapeutic class: Oral antidiabetic, Biguanides

Indication: Type II DM

Action: Decrease hepatic glucose production, decrease GI glucose absorption, increase cells sensitivity to insulin → **lower BG**

Nursing Considerations:

- Do not give to clients with renal or hepatic impairment
- Monitor
 - BG
 - BUN, Cr
- Take with food
- **Hold for 24 before and 48 hours after any study with IV contrast**

Part III: Endocrine Disorders

Hormone	Disease
<u>Steroids:</u> Glucocorticoids (cortisol) Mineralocorticoids (aldosterone)	<u>Not enough:</u> Addison's Disease
	<u>Too much:</u> Cushing's Disease

Addison's Disease

- Adrenocortical insufficiency - not enough steroids
- Decreased glucocorticoids
 - Fatigue
 - Weight loss
 - Hypoglycemia
 - Confusion
- Decreased mineralocorticoids
 - Loss of sodium and water → hyponatremic, fluid volume deficit
 - Retention of potassium → hyperkalemic
 - Hypotension
- Decreased sex hormones

Assessment

Hair loss

Hyperpigmentation

Weight loss

GI disturbances

Fatigue

Hypoglycemia



Postural hypotension

Weakness

Adrenal Crisis

Extreme fatigue

Dehydration

Fever

Hypotension

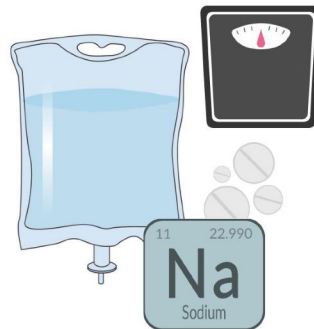
Renal shut down

↑ serum K

↓ serum NA

Treatment

- Think **SHOCK!**
 - IV fluid administration
 - Increased sodium intake
- I&O
- Daily weight
- Replace steroids
 - Prednisolone
 - Fludrocortisone



Cushing's Disease

- Excess of steroids
- Body has too much glucocorticoids, mineralocorticoids and sex hormones
 - Glucocorticoids
 - Immunosuppression
 - Hyperglycemia
 - Mood alteration
 - Fat redistribution (excess glucocorticoids cause lipolysis of extremities and lipogenesis in the trunk)
 - Mineralocorticoids
 - Fluid retention
 - Sodium retention
 - Potassium excretion
 - Sex hormones
 - Oily skin/acne

Cushing's Disease Assessment



Treatment

- Adrenalectomy
 - Remove the glands secreting the excess hormones
 - Can remove one or both
- **Avoid infection**
 - client is immunosuppressed
 - Hand washing
 - Limiting visitors



NCLEX Question

A nurse knows that in the event of an Addisonian crisis, it is most appropriate to administer which of the following medications intravenously?

- a. Insulin
- b. Normal saline solution
- c. dextrose 5% in water
- d. dextrose 5% in half-normal saline solution

Answer: B

One problem of a client in the Addisonian crisis is hyponatremia. The nurse should, therefore, anticipate administering the standard saline solution. Glucose, vasopressors, and hydrocortisone are also used to treat the Addisonian crisis. It would be inappropriate to administer insulin, dextrose 5% in water, or dextrose 5% in half-normal saline solution for this client. The correct answer is option B, while options A, C, and D are incorrect.

Conn's Disease

- “Primary aldosteronism”
- Caused by a tumor on the adrenal gland
 - Tumor produces too much aldosterone
 - Excessive production of specifically **aldosterone** by the adrenal context.
- Symptoms
 - Hypertension
 - Hypokalemia
 - Hypernatremia
- Treatment
 - Removal of tumor

Hormone	Disease
<u>Catecholamines:</u> Epinephrine Norepinephrine	<u>Not enough:</u> -
	<u>Too much:</u> Pheochromocytoma

Pheochromocytoma

- Tumor on the adrenal gland
 - Adrenal gland is responsible for catecholamine (epinephrine, norepinephrine) secretion
- Tumor causes excess secretion of catecholamines
- Avoid palpating the abdomen
 - Can stimulate catecholamine release

Assessment and treatment

Assessment

- Tachycardia
- Palpitations
- Hypertension
- Diaphoresis
- Abdominal pain
- Chest pain
- Severe headache

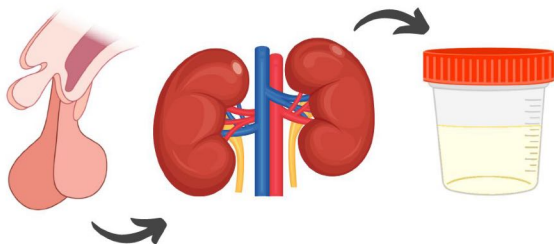
Treatment

- Alpha- and beta-adrenergic blockers for hypertension
- Surgery
 - Adrenalectomy

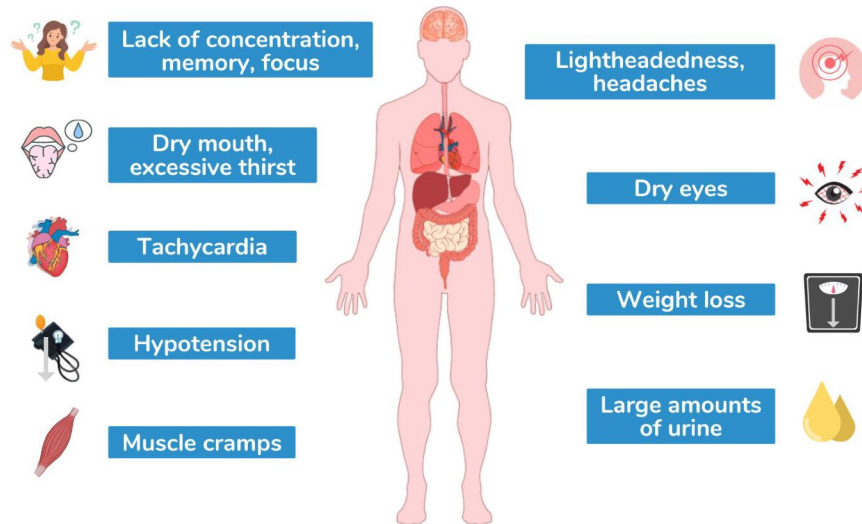
Hormone	Disease
<u>Antidiuretic Hormone:</u> ADH Vasopressin	<u>Not enough:</u> Diabetes Insipidus (DI)
	<u>Too much:</u> Syndrome of Inappropriate Antidiuretic Hormone (SIADH)

Diabetes Insipidus

- There is not enough ADH in the body
- Without ADH to tell the body to hold onto water, the kidneys produce HUGE amounts of urine.
- This leads to fluid volume deficit
- Hypotension
- Shock



DI Assessment



Lab Values

- Urine = dilute
 - Decreased USG
 - Decreased urine osmolarity
- Blood = concentrated
 - Increased Serum Na
 - Increased serum osmolarity
 - Serum Hct > 40%

Treatment

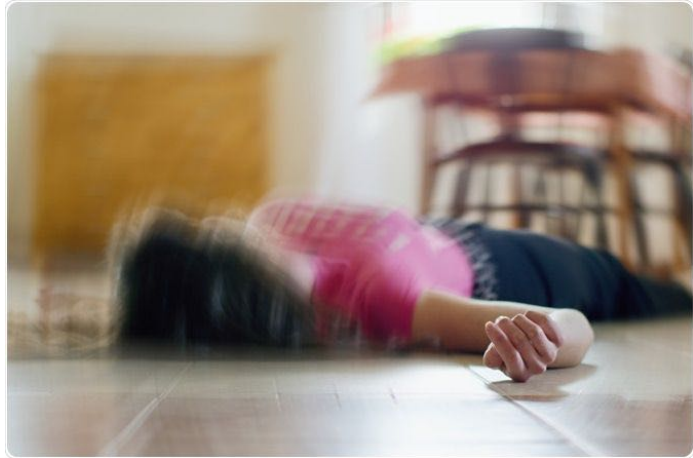
- **Monitor Neuro status**
- Replace fluids
 - Monitor hourly UOP
 - Replace volume + MIVF
- Vasopressin

Syndrome of Inappropriate Antidiuretic Hormone (SIDAH)

- The body is making too much ADH
- With too much *antidiuresis*, the kidneys stop excreting water and HOLD ON to it!
- Decreased UOP
- Retention of water in the intravascular space.
- ONLY water is retained.... No sodium.
 - Body remains euvolemic.

Assessment

- Weight gain
- NO peripheral edema
- Anorexia
- Nausea
- Vomiting
- Low serum sodium
 - Irritability
 - Confusion
 - Hallucinations
 - Seizures (Na < 125)



Lab values

- Urine = concentrated
 - Decreased UOP
 - Increased urine osmolality
 - Increased urine specific gravity
 - Increased urine sodium
- Blood = dilute
 - Increased blood volume
 - Decreased blood osmolality
 - Hyponatremia
 - Anemia

Treatment

- Monitor serum sodium
 - Sodium replacement
- Seizure precautions
- Fluid restriction
- Hypertonic saline

DI vs. SIADH

	DI	SIADH
Urine Output	High	Low
Urine Osmolarity	Low	High
Urine Specific Gravity	Low	High
Serum Sodium	High	Low
Serum Specific Gravity	High	Low

NCLEX Question

A client suddenly develops syndrome of inappropriate antidiuretic hormone (SIADH) after undergoing cranial surgery. Which manifestations should the nurse expect to see from the client?

- a. Edema and weight gain
- b. Decreased urine production
- c. Hypotension
- d. A low urine specific gravity

Answers: B

A is incorrect. Because of free water retention, there is increased retention of water in the intravascular space. But triggers the kidneys to excrete sodium, balancing the fluid status of the client. They are euvolemic. Weight gain is seen, but peripheral edema is not. This absence of peripheral edema is a specific finding of SIADH.

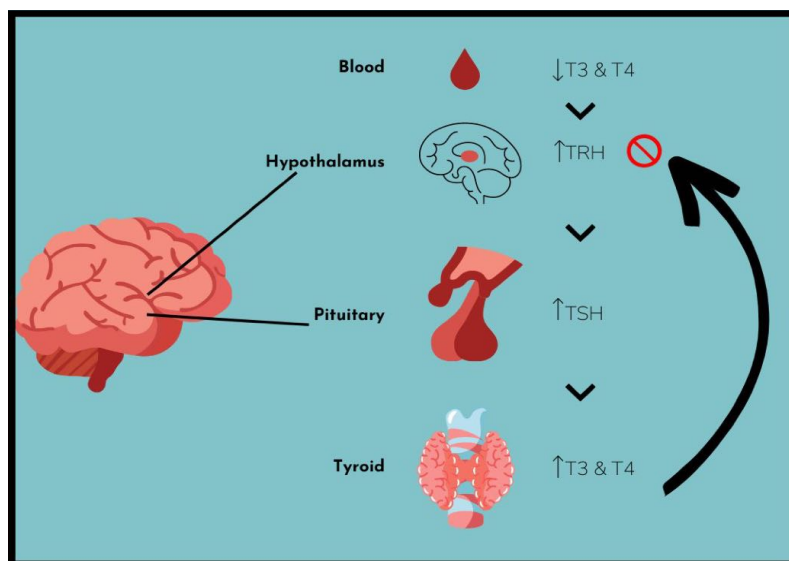
B is correct. Decreased urine production is a finding of SIADH. Because of the increase in ADH, there is an increased retention of free water and a decrease in urine output.

C is incorrect. You would expect to see a normal blood pressure in SIADH. Only free water is retained, no sodium, and the body remains in a euvolemic state. This means that clients with SIADH are normotensive.

D is incorrect. A low urine specific gravity would be seen in DI, when the production of ADH is decreased and the body secretes large amounts of dilute urine. In SIADH, the body retains free water and makes small amounts of concentrated urine, so the specific gravity is increased.

Hormone	Disease
<u>Thyroid Hormones:</u> TSH T3 T4	<u>Not enough:</u> Hypothyroidism
	<u>Too much:</u> Hyperthyroidism (Grave's Disease)

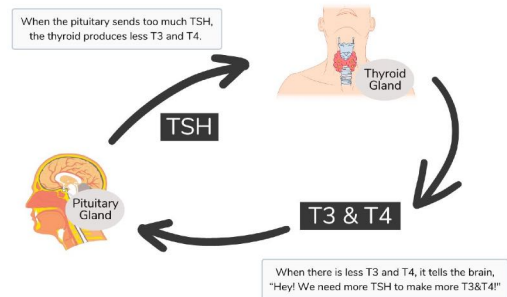
Thyroid hormone



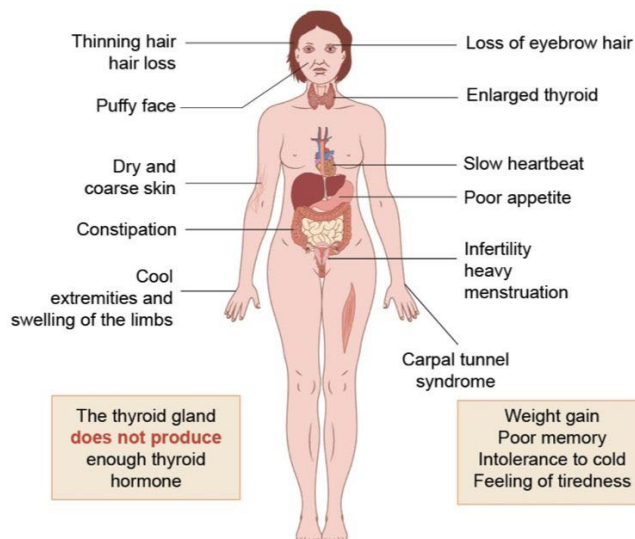
Hypothyroidism

- The body does not have enough thyroid hormone
- Increased levels of TSH trying to signal the thyroid to make more T3 and T4
- Thyroid gland cannot secrete enough T3 and T4 despite high TSH
- T3 and T4 continue to be low
- The negative feedback loop is broken

Low T4 / High TSH



Symptoms of Hypothyroidism



Treatment

- **Levothyroxine** - thyroid hormone
 - Take on an empty stomach
 - Take at the same time every day
 - Will take this forever

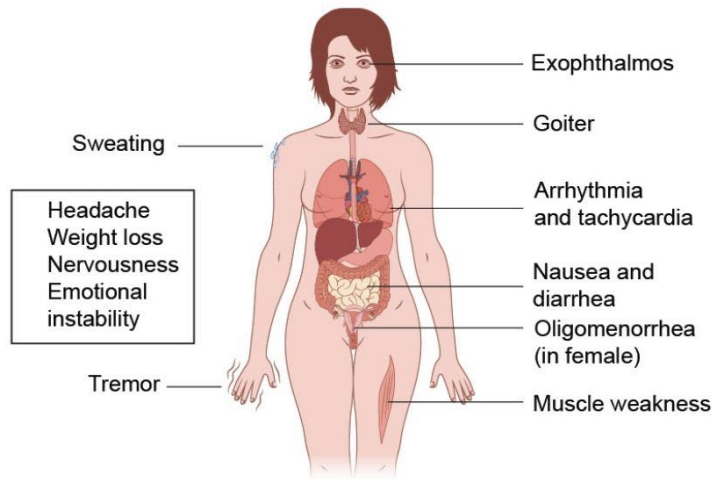


Hyperthyroidism

- Thyrotoxicosis - increased level of thyroid hormone
- Graves Disease - Autoimmune disease with antibodies attacking the thyroid
- ***The body has too much thyroid hormone!***
 - Decreased levels of TSH
 - Anterior pituitary sees low TSH and signals to the Thyroid gland to secrete more T3 and T4
 - T3 and T4 continue to be secreted despite being high
 - The negative feedback loop is broken

High T4 / Low TSH

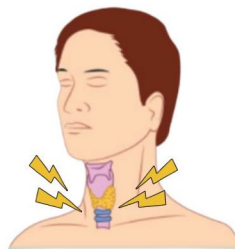
Graves disease symptoms



Thyroid Storm

Signs & Symptoms

-  Very high fever (100 to 106°F)
- Very high heart rate (up to 200 BPM) 
- Palpitations
-  Chest pain
- Shortness of breath 



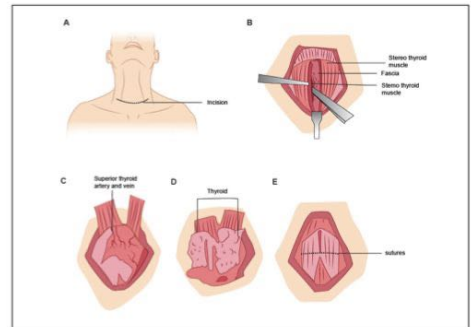
Risk Factors

- Untreated Graves' disease and/or hyperthyroidism 
- Recent surgery on thyroid gland or trauma to thyroid 

Treatment

- Antithyroid - methimazole
 - Stops the thyroid from making T3 and T4
- Iodine compounds
 - Used to decrease the size and vascularity of the thyroid gland
- Radioactive Iodine therapy
 - Destroys thyroid cells
 - Can cause hypothyroidism
- Thyroidectomy
 - Removal of all or some of the thyroid gland
 - If all of the thyroid gland is removed, replacement of thyroid hormone will be necessary post-op.

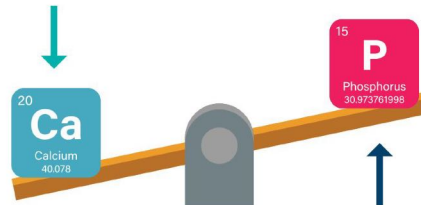
Thyroidectomy



Hormone	Disease
<u>Parathyroid Hormone</u>	<u>Not enough:</u> Hypoparathyroidism
	<u>Too much:</u> Hyperparathyroidism

Hypoparathyroidism

- The parathyroid glands do not secrete enough PTH
- There are low serum calcium levels
- Low serum calcium levels cause high serum phosphorus levels



Hypoparathyroidism Assessment

CNS

- Seizures
- Calcifications
- Parkinsonism or dystonia

PNS

- Paresthesia
- Muscle cramps
- Tetany

Cardiovascular

- Cardiac arrhythmias
- Hypocalcemia-associated dilated cardiomyopathy

Musculoskeletal

- Myopathy
- Spondyloarthropathy

Respiratory

- Laryngospasm

Neuropsychiatric

- Anxiety & depression

Renal

- Nephrocalcinosis
- Kidney stones
- Chronic kidney disease

Dermatological

- Dry skin
- Onycholysis
- Course, thin hair
- Pustular psoriasis

Ophthalmological

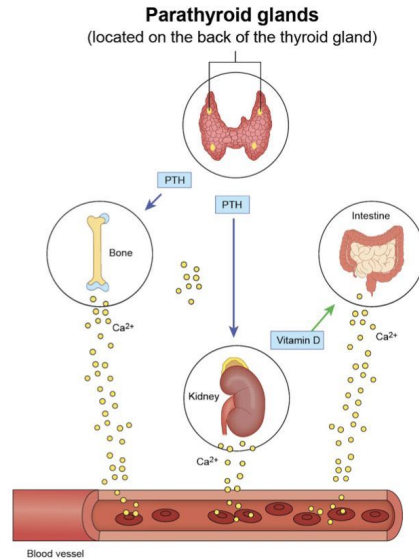
- Cataracts
- Papilledema

Dental

- Altered tooth morphology

Treatment

- Fix the electrolyte imbalances
 - Calcium replacement
 - Phosphorus binders



Hyperparathyroidism

- The parathyroid glands secrete too much PTH
- There are high serum calcium levels
- High serum calcium levels cause low serum phosphorus levels



Hyperparathyroidism Assessment

Digestive

- Loss of appetite
- Nausea
- Vomiting
- Constipation



Nervous System

- Fatigue
- Depression
- Confusion



Musculoskeletal

- Bone/joint aches & pains
- Muscle weakness



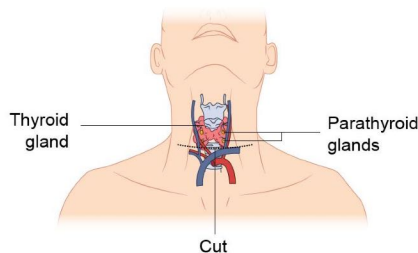
Urinary

- Kidney stones
- Increased thirst
- Increased urination

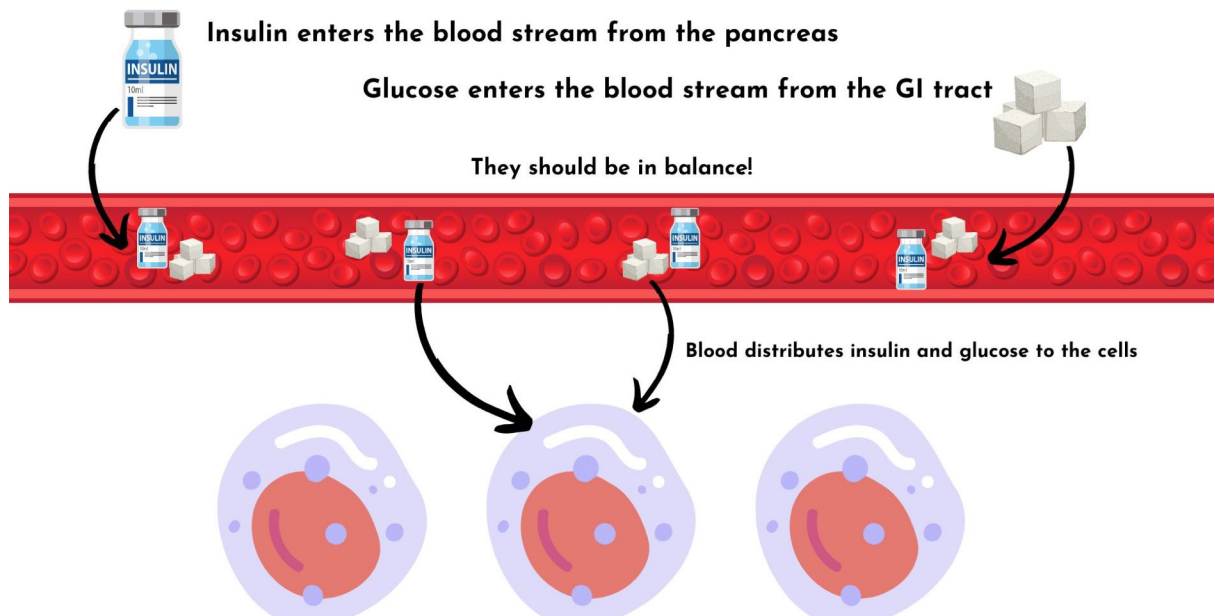


Treatment

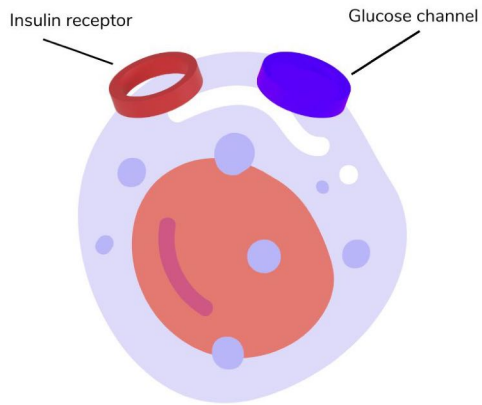
- Partial parathyroidectomy
 - There are 4 parathyroid glands
 - Taking out 2 can decrease PTH secretion
 - Can cause rebound hypocalcemia if decreases too much



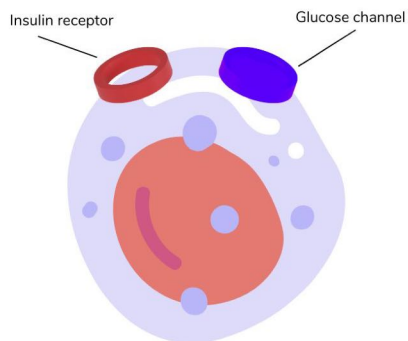
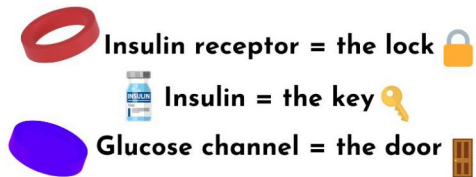
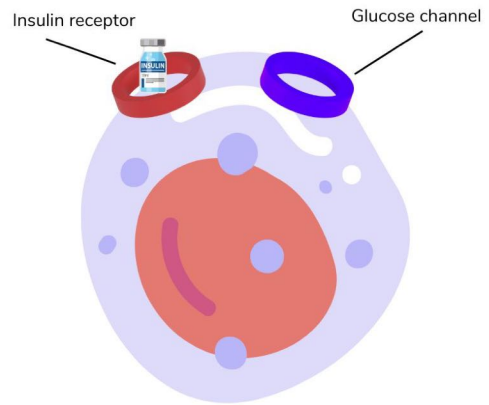
Hormone	Disease
<u>Insulin</u>	<u>Not enough:</u> Diabetes: Type I, Type II
	<u>Too much:</u> Hypoglycemia



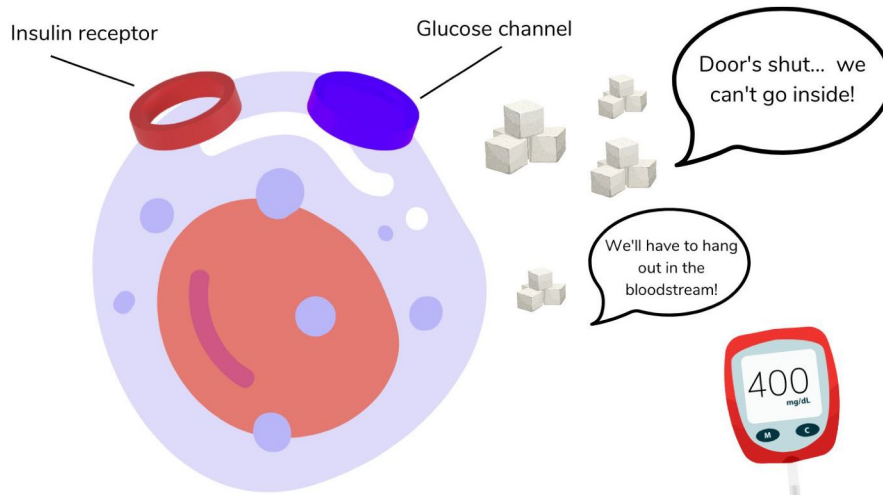
**No Insulin -
glucose channel is closed**



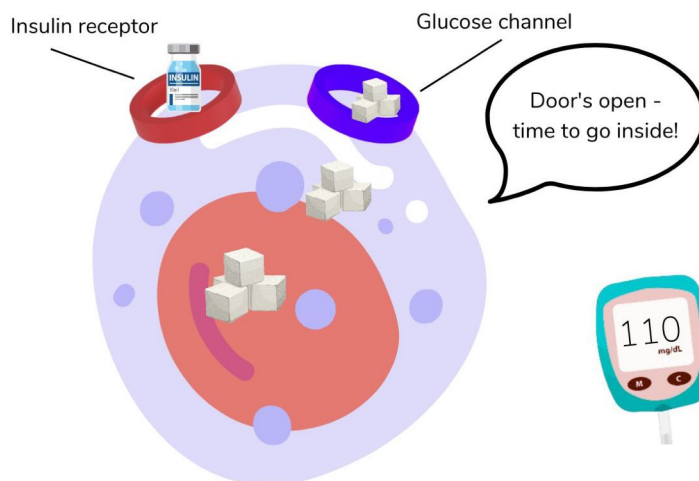
**Insulin -
glucose channel is open**



No Insulin.... No key.... Door is shut!



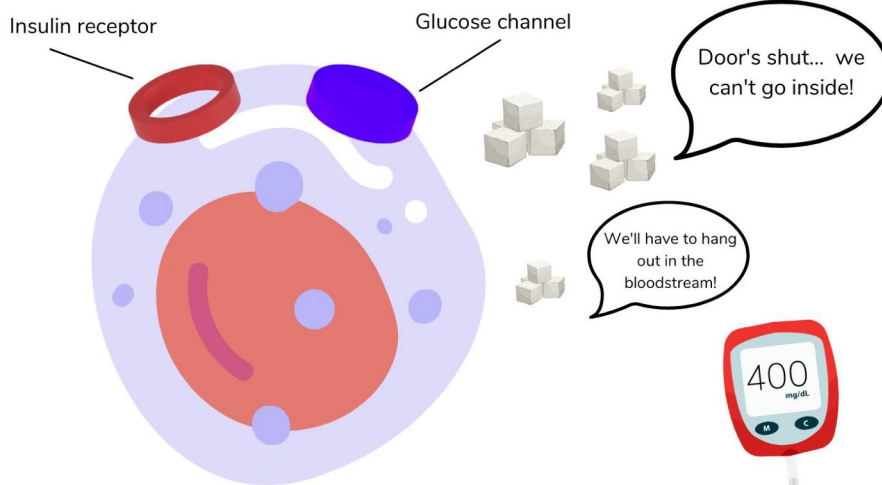
WITH Insulin... key unlocks the door.... glucose goes inside!



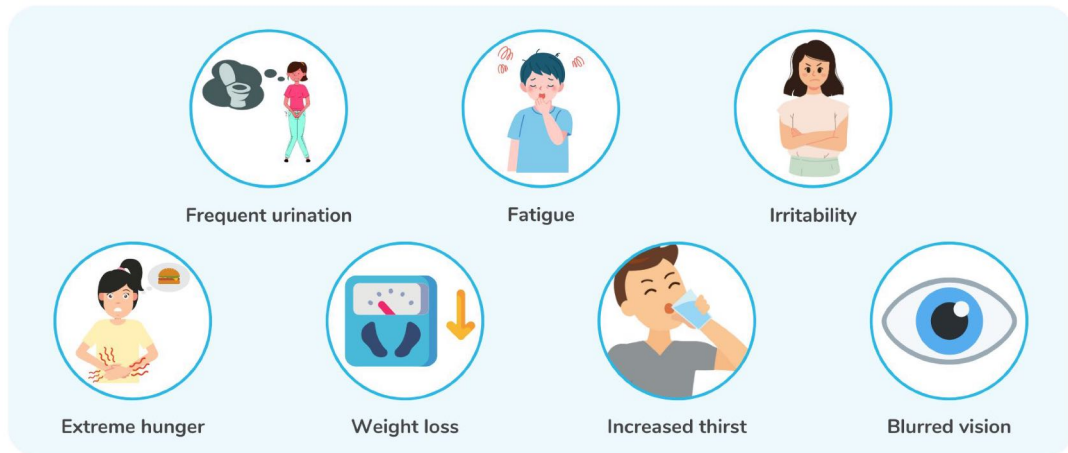
Diabetes Mellitus Type I

- DMTI
- Autoimmune disease - or idiopathic
- Body has destroyed the beta cells of the pancreas that produce insulin
- There is little or no insulin in the body
- Very high levels of glucose in the bloodstream
- No glucose can get to the cells for fuel

No Insulin.... No key.... Door is shut!



Diabetes Mellitus Type I Assessment



Diabetic Ketoacidosis

- There is no insulin to carry glucose to the cells
- Glucose builds up in the blood (High BG)
- Blood becomes hypertonic, causing fluid to shift into the vascular space.
- Kidneys work to filter this excess fluid and glucose - polyuria
- Cells are not receiving any fluid or glucose - they are starving - polydipsia & polyphagia
- Because cells don't have any glucose for energy, break down proteins and fat
- This produces ketones - which are an acid
- Causes a metabolic acidosis
 - Kussmaul respirations - to blow off CO₂ to compensate
 - High serum potassium

DKA Assessment

Early Signs

Feeling very thirsty



Urinary frequency



High blood
glucose levels

High ketone levels
in urine



Polyuria
Polydipsia
Polyphagia

Later/Severe Signs



Feeling weak or
constantly sleepy

Dry, flushed skin

Nausea, vomiting,
abdominal pain

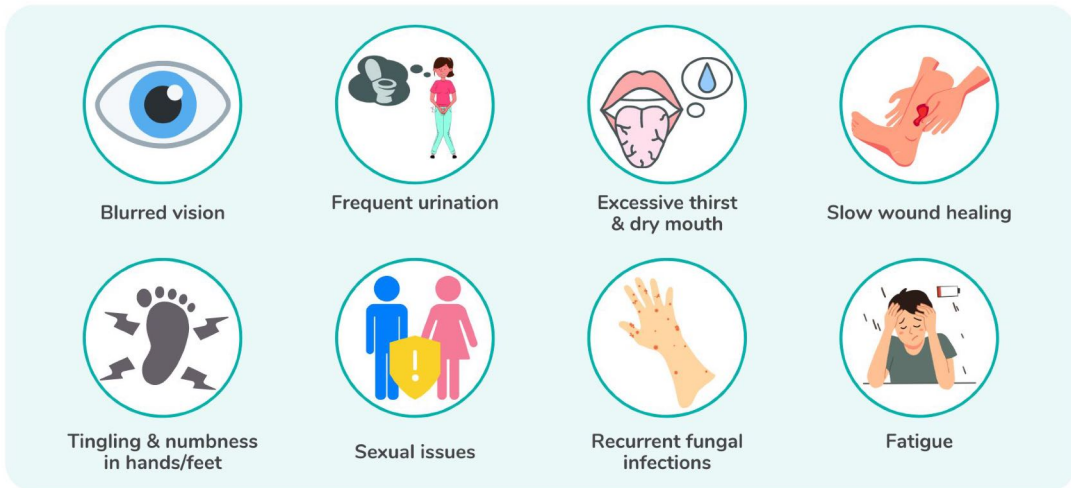


Difficulty breathing &
fruity-smelling breath

Diabetes Mellitus Type II

- There is either not enough insulin, insulin resistance, or bad insulin
- Commonly found with clients who are overweight.
- Their body can't make enough insulin to keep up with the glucose.

Diabetes Mellitus Type II Assessment



Hyperglycemic Hyperosmolar **Nonketotic** Syndrome (HHNS)

- Exacerbation of T1DM
- Pathophysiology:
 - Extremely high blood glucose levels
 - Blood becomes hyperosmolar
 - Kidneys start producing more urine due to the hyperosmolarity of blood
 - Polyuria → dehydration → Shock
 - **No ketones**
- Symptoms begin over days to weeks (vs. hours for DKA)



Type I vs. Type II DM:

KetONE bodies are seen
in type ONE diabetes



Treatment - Acute

- Insulin
 - *Move that glucose back into the cell!*
- IV Fluids
 - Restore intravascular volume
 - Remember - high BG caused a fluid shift out of the vascular space!
- Electrolyte replacement
 - Potassium

Treatment - Long Term

DIET

- Low carb - complex carbs
- Proteins & veggies

EXERCISE

- Eat before exercising
- Exercise when blood sugar is at its highest
- Establish a routine

ORAL AGENTS

- Work to decrease the amount of circulating glucose
- Improves how the body produces insulin and uses insulin
- Metformin

INSULIN

- Basal bolus system
- Long-acting agent given once per day
- Rapid-acting agent given with meals to cover the carbs eaten

Hypoglycemia

- When there is not enough glucose in the bloodstream
- BG <70
- Causes
 - Not enough food
 - Too much insulin
 - Too much exercise

Hypoglycemia Signs & Symptoms "TIRED"



Tachycardia
Irritability
Restlessness
Excessive hunger
Diaphoresis

Treatment

1. Have a snack - about **15 grams of carbs**
 - a. 4-6 oz of soda/juice/milk
 - b. 8-10 pieces of candy
2. Wait **15 minutes**, and check BG again
3. If still <70, eat another **15 grams** of carbs
4. After the BG rises, eat a snack with complex carb/protein to help keep the BG up
 - a. Crackers with peanut butter

15-15-15

What if the client is unconscious?!

If IV access → push D50W

If no IV → IM Glucagon (catabolic hormone, raises concentration of glucose in the bloodstream)

Questions for our tutors? nclextutors@archerreview.com

Small Group Tutoring

Endocrine Disorders

Fluids + Electrolytes 101

Focus on practice questions and learn what you need to do to get them correct!!

Scan the QR code to sign up!

